References

- Ideals generated by R-sequences - Vasconcelos
- Vasconcelos' conjecture on the conormal module - Benjamin Briggs
- When is an ideal gen. by a regular sequence? - M. Boratyński

CAAGS - 2026

When is an ideal generated by a regular seq.?

Wolmer Vasconcelos - 1966

We will work with Noetherian local rings.

Let $(R, \mathfrak{m})$. Given $x_1 \dots x_n \in R$ is said to be a regular seq. if x_i is regular on $\frac{R}{(x_1 \dots x_{i-1})}$

$\forall \ 1 \leq i \leq n$

- It is "ideal" to have an ided gen by a regular sequence.

- Let I be gen. by a regular seq. Then $\frac{I}{I^2}$ is free $\frac{R}{I}$-module. (Ex)

Ques: Is it sufficient? - No!

- $R = \frac{k[[x,y]]}{(xy)}$ $\quad I = \mathfrak{m} = (\bar{x}, \bar{y})$

$\frac{I}{I^2} = \frac{\mathfrak{m}}{\mathfrak{m}\cdot\mathfrak{m}}$ is free. $\mu(\mathfrak{m}) = \dim_k\left(\frac{\mathfrak{m}}{\mathfrak{m}^2}\right) = 2$

If $\mathfrak{m}$ was gen. by a regular seq, then

$$\operatorname{ht}\mathfrak{m} = \mu(\mathfrak{m}) = 2$$
$$\| \quad 1$$

which is not possible!

- $R = \frac{k[[x]]}{(x^2)}$ $\quad I = (\bar{x})$

$\frac{I}{I^2} \cong \frac{R}{I}$. So it is free but I is not gen. by a regular seq.

Ques: What is missing here?

- I is gen. by a regular seq $\Rightarrow \operatorname{pdim}_R(I) < \infty$

Is it sufficient? **No**

$R = k[[x,y]]$ $\quad I = (x^2, xy)$

$0 \longrightarrow R \xrightarrow{\begin{pmatrix} -y \\ x \end{pmatrix}} R^2 \xrightarrow{(x^2 \; xy)} I \longrightarrow 0$ is exact

Hence $\operatorname{pdim}(I) = 1 < \infty$.

Is it gen. by a regular sequence?

$\sqrt{I} = (x) \Rightarrow ht\, I = 1$

$\mu(I) = 2$. So not gen. by a regular seq?

$R = \dfrac{k[[x,y]]}{(xy)}$ $\quad \mathfrak{m} = (\bar{x}, \bar{y})$

$\operatorname{pdim}(\mathfrak{m}) = \infty$. If not, $\operatorname{pdim}(\mathfrak{m}) + \operatorname{depth}_R \mathfrak{m} = \dim R$.

$\operatorname{depth}_R \mathfrak{m} = \dim R = 1 \Rightarrow \operatorname{pdim}(\mathfrak{m}) = 0$

So $\mathfrak{m}$ is free R-module which is not possible.

If $R = k[[x,y]]$ & $I = (x^2, xy)$.

Can $\dfrac{I}{I^2}$ be free? $\quad \mu\left(\dfrac{I}{I^2}\right) = 2$

Infact $\dfrac{I}{I^2}$ has rank 2 over k.

Consider $\left(\dfrac{R}{I}\right)^2 \longrightarrow \dfrac{I}{I^2} \xrightarrow{\sim} \left(\dfrac{R}{I}\right)^2$

$(\bar{a}, \bar{b}) \longmapsto ax^2 + bxy + I^2$.

This is surjective but not injective.

Thm: Let $(R, \mathfrak{m})$ be a Noeth. local ring & I be an ideal of finite projective dim. Suppose $\exists$ an ideal J s.t $I^2 \subseteq J \subseteq I$ and that $\frac{I}{J}$ is a free $\frac{R}{I}$-module. Then $\exists$ R-regular sq $x_1, \dots, x_r$ in I such that $I = (x_1, \dots, x_n) + J$.

Moreover, if $J \subseteq \mathfrak{m}I$, then $I = (x_1, \dots x_r)$.

$I = (x_1, \dots x_n) + J$, if $J \subseteq \mathfrak{m}I$

$\Rightarrow I = (x_1, \dots x_n) + J \subseteq (x_1, \dots x_n) + \mathfrak{m}I \subseteq I$

$\Rightarrow I = (x_1, \dots x_n) + \mathfrak{m}I$

By NAK, $I = (x_1, \dots x_n)$.

In particular, take $J = I^2$.

If $\operatorname{pdim} I < \infty$, $\frac{I}{I^2}$ is free, then $I^2 \subseteq \mathfrak{m}I$

& hence $I = (x_1, \dots x_r)$.

what does $\frac{I}{J}$ being free means?

$\Rightarrow \exists$ $x_1, \dots x_k \in I$ s.t

$I = (x_1, \dots x_k) + J$ and whenever

$a_1 x_1 + \dots + a_k x_k \in J \Rightarrow a_i \in I \ \forall \ i$.

Possible difficulty / strategy to prove this!

- Can we pick a nzd in I, say x_1 which is part of a basis of $\frac{I}{J}$?
- After passing to $\frac{R}{(x_1)}$, will the freeness & finiteness hypothesis remains.
- Want to drop rank from r to $r-1$.

PROOF:) $\frac{I}{J}$ is free, $\operatorname{pdim}(I) < \infty$, $r = \operatorname{rank}\left(\frac{I}{J}\right)$.

Take $r=0$, $I = J$, so empty regular seq works.

Assume that $\left(\frac{I}{J}\right) \cong \left(\frac{R}{I}\right)^r$ & the result holds for rank $r-1$.

- Every non-zero ideal having a finite proj. dimension contains a nzd. This follows from the following:

Lemma: If R is a ring & M is a fg R-module with $0 \to F_r \to \cdots \to F_0 \to M \to 0$

Then TFAE
1. $\operatorname{Ann}_R(M) \neq 0$
2. $\operatorname{Ann}_R(M)$ contains a nzd in R.

Take $M = \frac{R}{I}$

$\exists$ a nzd of R in I say x_1.

Lemma: For a local ring $(A, \mathfrak{m})$ & a free module $N \cong A^n$, tf.ae. for $v \in N$.

1) v can be extended to an A-basis of N.

2) $v \notin \mathfrak{m}N$

Apply this to $A = \frac{R}{I}$ $\quad N = \frac{I}{J}$ $\quad \mathfrak{m} = \frac{\mathfrak{M}}{I}$

$$x_1 + J \notin \mathfrak{m}N = \frac{\mathfrak{M}I + J}{J} \iff x_1 \in I \setminus \mathfrak{M}I + J.$$

Suppose not. Then every element of $I \setminus \mathfrak{M}I + J$ is a zero divisor in R.

$$\Rightarrow \quad I \setminus \mathfrak{M}I + J \subseteq \bigcup_{i=1}^{s} P_i.$$

Pick $y \in I \setminus \mathfrak{M}I + J$ & any $x \in \mathfrak{M}I + J$.

$y - x^n$, $n \in \mathbb{N}$. Note $y - x^n \in I \setminus \mathfrak{M}I + J \ \forall n$.

$$\Rightarrow \quad y - x^n \in \bigcup P_i \quad \forall n.$$

$\Rightarrow \exists\ i > j$ s.t $\quad y - x^i \in P_k$ & $y - x^j \in P_k$

$$\Rightarrow \quad x^j(x^{i-j} - 1) \in P_k \Rightarrow x^j \in P_k \Rightarrow x \in P_k.$$

$$\Rightarrow \quad \mathfrak{M}I + J \subseteq \bigcup P_i$$

Combining with $I \setminus \mathfrak{M}I + J \subseteq \bigcup P_i$

$\Rightarrow I \subseteq \bigcup P_i \Rightarrow$ Every element of I is a zero divisor in R

Thus, $\exists\ x_2, \dots x_r$ s.t. $x_1+J, \dots, x_r+J$ is an $\frac{R}{I}$-basis of $\frac{I}{J}$.

$\Rightarrow I = (x_1, \dots x_r) + J$ where x_1 is nzd on R.

Take $R^* = \frac{R}{(x_1)}$, $I^* = \frac{I}{(x_1)}$ $J^* = \frac{J+(x_1)}{(x_1)}$.

$I^{*2} \subseteq J^* \subseteq I$ ✓

$\frac{I}{J}$ is free of rank r.

$\frac{I^*}{J^*} \cong \left(\frac{R^*}{I^*}\right)^{r-1}$ since $x_2^*, \dots x_r^*$ forms an $\frac{R^*}{I^*}$ basis of $\frac{I^*}{J^*}$

Remains to prove $\operatorname{pdim}_{R^*}\left(\frac{I}{(x_1)}\right) < \infty$.

Recall $\operatorname{pd}_{\frac{R}{x_1}}\left(\frac{M}{x_1 M}\right) = \operatorname{pd} M$ if x is a nzd on both M & R.

$\Rightarrow \operatorname{pdim}\left(\frac{I}{x_1 I}\right) = \operatorname{pdim}\left(\frac{I}{x_1 R}\right) < \infty$.

We want $\operatorname{pdim}\left(\frac{I}{x_1 R}\right) < \infty$.

Lemma: $\frac{I}{x_1 I} \cong \frac{(x_1)}{x_1 I} \oplus \frac{I}{x_1 R}$

Thus $\operatorname{pdim}\left(\frac{I}{x_1R}\right) = \operatorname{pdim}(I^*) < \infty$.

By induction, $\exists\ y_2^*, \dots y_r^* \in I^*$ which forms a regular seq in $\frac{R}{(x_1)}$ s.t

$$I^* = (y_2^*, \dots y_r^*) + J^*.$$

$\Rightarrow (x_1, y_2, \dots y_r)$ is R-regular. &

$$I = (x_1, y_2, \dots y_r) + J$$

□

- If I is an $\mathfrak{M}$-primary ideal and $\frac{I}{I^2}$ is free Then $\mu(I) \leq \mu(\mathfrak{M})$.

FINAL REMARKS

- Benjamin Briggs in 2022 proved that instead of requiring $\frac{I}{I^2}$ to be free, it is enough to conclude when $\operatorname{pdim}\left(\frac{I}{I^2}\right) < \infty$.

Used methods from Homotopy theory.

- Vasconcelos conjectured the following.

If R is any Noetherian ring for which projectives are free & I is an ideal st $\frac{I}{I^2}$ is free, $\operatorname{pdim}(I) < \infty$, then

I is gen. by a regular seq.

In 1979, it was disproved by Boratynski.

Is the conjecture true if $R = k[x_1, \dots, x_n]$?

- The importance of Co-normal module $\frac{I}{I^2}$?